

torch.cholesky_solve#

https://pytorch.org/docs/stable/generated/torch.cholesky_solve.html#torch.cholesky_solve

Description:

Computes the solution of a system of linear equations with complex Hermitian or real symmetric positive-definite lhs given its Cholesky decomposition. Let AAA be a complex Hermitian or real symmetric positive-definite matrix, and LLL its Cholesky decomposition such that: where LHL^{H} is the conjugate transpose when LLL is complex, and the transpose when LLL is real-valued. Returns the solution XXX of the following linear system: Supports inputs of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if AAA or BBB is a batch of matrices then the output has the same batch dimensions.

Function Signature

```
torch.cholesky_solve(B, L, upper=False, *, out=None) →  
Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
>>> A = torch.randn(3, 3)
>>> A = A @ A.T + torch.eye(3) * 1e-3 # Creates a symmetric
positive-definite matrix
>>> L = torch.linalg.cholesky(A) # Extract Cholesky
decomposition
>>> B = torch.randn(3, 2)
>>> torch.cholesky_solve(B, L)
tensor([[ -8.1625,  19.6097],
        [ -5.8398,  14.2387],
        [ -4.3771,  10.4173]])
>>> A.inverse() @ B
tensor([[ -8.1626,  19.6097],
        [ -5.8398,  14.2387],
        [ -4.3771,  10.4173]])

>>> A = torch.randn(3, 2, 2, dtype=torch.complex64)
>>> A = A @ A.mH + torch.eye(2) * 1e-3 # Batch of Hermitian
positive-definite matrices
>>> L = torch.linalg.cholesky(A)
>>> B = torch.randn(2, 1, dtype=torch.complex64)
>>> X = torch.cholesky_solve(B, L)
>>> torch.dist(X, A.inverse() @ B)
tensor(1.6881e-5)
```

Detailed Documentation

Documentation

Computes the solution of a system of linear equations with complex Hermitian or real symmetric positive-definite lhs given its Cholesky decomposition.

Let AAA be a complex Hermitian or real symmetric positive-definite matrix, and LLL its Cholesky decomposition such that:

where LHL^{H} is the conjugate transpose when LLL is complex, and the transpose when LLL is real-valued.

Returns the solution XXX of the following linear system:

Supports inputs of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if AAA or BBB is a batch of matrices then the output has the same batch dimensions.

B (Tensor) – right-hand side tensor of shape $(*, n, k)$ where $*$ is zero or more batch dimensions

L (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions consisting of lower or upper triangular Cholesky decompositions of symmetric or Hermitian positive-definite matrices.

`upper` (bool, optional) – flag that indicates whether LLL is lower triangular or upper triangular. Default: False.

`out` (Tensor, optional) – output tensor. Ignored if None. Default: None.

